In [1]:
```
# 1. E-commerce Discount Calculator
# Scenario: An online store offers discounts based on the purchase amount:
# 10% discount for purchases between $100 and $500
# 20% discount for purchases above $500
# No discount for purchases below $100
# Task: Write a program that takes the purchase amount as input and calculates t
pa=float(input("Enter the purchase amount:"))
if pa>=100 and pa<=500:
    print("you got 10% discount:")
    dis=pa*10/100
elif pa>500:
    print("you got 20% discount:")
    dis=pa*20/100
elif pa<100:
    print("no discount")
    dis=0
print(f"The discount amount is:{dis}\nThe total bill amount is{pa-dis}:")
```

```
you got 20% discount:
The discount amount is:120.0
The total bill amount is480.0:
```

In [2]:
```
# 2. Traffic Light Simulation
# Scenario: Create a program that simulates a traffic light. The user inputs one
# If the input is Red, display "Stop."
# If the input is Yellow, display "Ready to move."
# If the input is Green, display "Go."
# For invalid input, display "Invalid color."
tl=input("enter the traffic light")
if tl=="Red":
    print("stop")
elif tl=="Yellow":
    print("Ready to move")
elif tl=="Green":
    print("Go")
else:
    print("invalid color")
```

```
stop
```

In [5]:
```
# 3. Grade Evaluation System
# Scenario: A school uses the following grading system:
# Marks >= 90: Grade A
# Marks >= 75 and < 90: Grade B
# Marks >= 50 and < 75: Grade C
# Marks < 50: Fail
# Task: Write a program that accepts the student's marks and displays their grad
per=float(input("enter a number:"))
if(per>=90 and per<=100):
    print("Grade A")
elif(per>=75 and per<90):
    print("Grade B")
elif(per>=50 and per<75):
    print("Grade C")
elif(per>=0 and per<50):
    print("fail")
else:
    print("Invalid input")
```

```
Grade A
```

In [9]:
```python
# 4. Odd or Even and Divisibility Check
# Scenario: Write a program that takes an integer as input and checks:
# Whether the number is odd or even
# Whether the number is divisible by 5
# Display appropriate messages for both conditions.
number=int(input("enter the number:"))
if(number%2==0):
    print("it is even number")
else:
    print("it is odd number")
if(number%5==0):
    print("the number is divisible by 5")
else:
    print("the number is not divisible by 5")
```

```
it is even number
the number is not divisible by 5
```

In [10]:
```python
# 5. Password Strength Checker
# Scenario: Write a program that checks the strength of a password based on thes
# Length >= 8 characters: Strong
# Length between 5 and 7 characters: Medium
# Length < 5 characters: Weak
pswd=input("Enter the Password:")
pswdlen=len(pswd)
if(pswdlen>=8):
    print("Strong password")
elif(pswdlen>=5 and pswdlen<=7):
    print("Medium password")
elif(pswdlen<5):
    print("Weak password")
```

```
Strong password
```

In [11]:
```python
# 6. Electricity Bill Calculator
# Scenario: An electricity company charges its customers as follows:
# First 100 units: $0.5 per unit
# Next 100 units (101-200): $0.75 per unit
# Above 200 units: $1 per unit
# Task: Write a program that accepts the number of units consumed and calculates
units=int(input("enter the electricity units:"))
if units<=100:
    bill=units+0.5
    print("electricity bill",bill)
elif units<=200:
    bill=(100*0.5)+((units-100)*0.75)
    print("electricity bill",bill)
else:
    bill=(100*0.5)+(100*0.75)+((units-200)*1)
    print("electricity bill",bill)
```

```
electricity bill 90.5
```

In [12]:
```python
# 7. Eligibility for Loan Approval
# Scenario: A bank approves loans based on these conditions:
# Age should be between 21 and 60
# Monthly income should be greater than or equal to $5000
# Task: Write a program to check whether a person is eligible for the loan based
age=int(input("enter the age:"))
```

```
mi=float(input("enter the monthly income"))
if(age<21 and age>60):
    print("not eligible for loan approval")
elif(mi<5000):
    print("not eligible for loan approval")
else:
    print("eligible for loan approval")
```

```
eligible for loan approval
```

In [13]:
```
# 8. Temperature Alert System
# Scenario: A weather monitoring system alerts based on the temperature:
# Below 0°C: "Freezing weather"
# 0°C to 20°C: "Cold weather"
# 21°C to 30°C: "Warm weather"
# Above 30°C: "Hot weather"
# Task: Write a program that takes the temperature as input and displays the cor
temp=float(input("enter the temperature"))
if temp<0:
    print("Freezing weather")
elif temp<=20:
    print("cold weather")
elif temp<=30:
    print("Warm weather")
else:
    print("hot weather")
```

```
Warm weather
```

In [ ]:
```

```